

* Home Assignment - 3 *

A)

~~3) The position vector~~4) For a particle with P.V. $r(t) = t^3 i + t^2 j + t k$ a) Find velocity $v(t)$ & acc. $a(t)$ at $t=1$.b) Determine angle betⁿ tangent vectors at $t=1$ & $t=3$.

$$a) \quad v(t) = 3t^2 i + 2t j + k$$

$$v(1) = 3i + 2j + k //$$

$$a(t) = 6t i + 2j + 0k$$

$$a(1) = 6i + 2j //$$

$$b) \quad v(3) = 27i + 6j + k$$

$$\therefore \cos \theta = \frac{v(1) \cdot v(3)}{|v(1)| |v(3)|} = \frac{3(27) + 12 + 1}{\sqrt{9+4+1} \sqrt{729+36+1}}$$

$$= \frac{94}{\sqrt{14} \sqrt{766}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{94}{\sqrt{14 \cdot 766}} \right) //$$

B) 8] Suppose you are climbing Hill whose shape is $z = 1000 - 0.005x^2 - 0.01y^2$, where x, y & z are measured in m. & you are standing at pt with coords $(60, 40, 966)$. +ve x -axis pts east & +ve y -axis pts. north.

a) If you start to walk towards south, will you start to ascend or descend? at what rate?

b) If you walk northwest — " — ?

a) gradient of z (∇z) $\Rightarrow$

$$\therefore \frac{\partial z}{\partial x} = -0.01x$$

$$\therefore \frac{\partial z}{\partial y} = -0.02y$$

$$\therefore \nabla z = \left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \right) = (-0.01x, -0.02y)$$

$\therefore$ at pt. $(60, 40)$,

$$\nabla z = (-0.6, -0.8)$$

For south, coords $\Rightarrow (0, -1)$

$$D_{\text{south}} z = \nabla z \cdot (0, -1) = 0.8$$

$\therefore$ ascend at rate of 0.8 m/m

b) For northwest, coords $\Rightarrow \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$

$$\therefore D_{\text{northwest}} z = \nabla z \cdot \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$$

$$= (-0.6, -0.8) \cdot \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$$

$$= 0.4242 - 0.5656$$

$$= -0.1414$$

$\therefore$ Descent at rate of -0.1414

$$\frac{0.707}{\sqrt{2}}$$

$$\frac{0.4242}{\sqrt{2}}$$

$$0.5656$$

c]

- 6] Velocity field of cyclone is $V = y^2z\mathbf{i} + z^2x\mathbf{j} + x^2y\mathbf{k}$.
Calculate curl to understand rotational behaviour.

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$$\nabla V = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2z & z^2x & x^2y \end{vmatrix}$$

$$= \mathbf{i} \left(\frac{\partial}{\partial y}(x^2y) - \frac{\partial}{\partial z}(z^2x) \right) - \mathbf{j} \left(\frac{\partial}{\partial x}(x^2y) - \frac{\partial}{\partial z}(y^2z) \right) \\ + \mathbf{k} \left(\frac{\partial}{\partial x}(z^2x) - \frac{\partial}{\partial y}(y^2z) \right)$$

$$\nabla V = \mathbf{i}(x^2 - 2zx) - \mathbf{j}(2xy - y^2) + \mathbf{k}(z^2 - 2yz)$$

→ Curl

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